

Stat Profile Prime® Critical Care Analyser

New consumer based micro-electronics and new technology micro-sensor cartridges are combined to produce a smaller, faster, more powerful, and less expensive critical care blood gas analyser.



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Stat Profile Prime Technology Delivers Exceptional Value

Stat Profile Prime combines the micro-electronics technology that revolutionized consumer electronics with Nova Biomedical's advanced micro-sensor technology. These technologies reduce the size, components, cost, weight, and maintenance of Stat Profile Prime, while improving speed, throughput, and uptime.

Stat Profile Prime makes critical care testing easy and affordable for any hospital, featuring:

- 10-test critical care menu
- Results in 60 seconds
- Fast throughput
- Zero maintenance
- 24-hour readiness
- True liquid quality control (QC)
- Low cost

10 Critical Care Tests, Results in 60 Seconds

Critical care testing requires instrumentation with a menu of essential tests to enable immediate diagnosis and treatment of critical illness. Equally important is 24-hour instrument readiness and rapid analysis time. Stat Profile Prime is uniquely designed to meet these needs by delivering a 10-test critical care profile in only 60 seconds.

10-test critical care menu

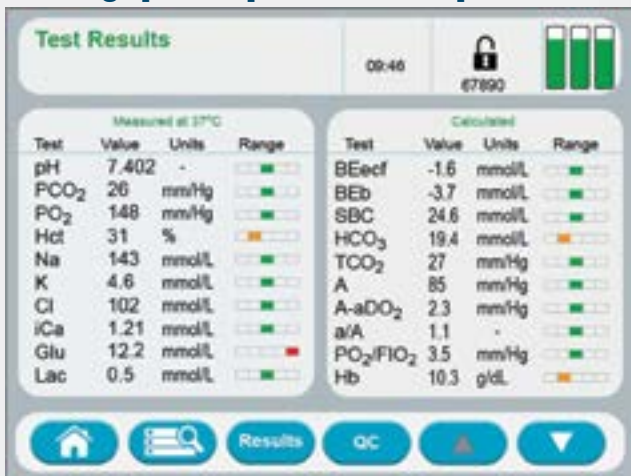
pH, PCO₂, PO₂, Na, K, Cl, iCa, Glu, Lac, Hct

Fast results in 60 seconds

Arterial, venous, or capillary micro-samples

- 100 microliters: full menu
- 50 microliters: blood gases

Throughput: Up to 45 Samples/Hour



results screen



Stat Profile Prime MicroSensor Technology



All sensors are contained in one MicroSensor Card. (actual size)

Accuracy

Stat Profile Prime uses proven Nova measurement technology in miniaturized, sensor card format.

Constant stat readiness

The MicroSensor Card has an on-board use life of 32 days, and is automatically calibrated and always ready for analysis.

Clot protection

Our unique Clot Block™ sample flow path is designed to protect sensor cartridges from blood clot blockages.

Simple, Fast Operation



home screen

Easy-to-use, high definition, color touchscreen

The touchscreen is easily operated through the use of simple and intuitive prompts and requires minimal training.

Testing in three simple steps

1. Press “Start” 2. Scan or enter patient ID 3. Press “Aspirate”

Integrated barcode scanner

An optional integrated 1D/2D barcode scanner conveniently located within the sample port eliminates external handheld scanners and allows for fast, error free entry of operator and patient IDs.

Easy sampling from syringes, capillaries, tubes, and ampoules

All testing uses a single sample port. Even capillary sampling is performed without adapters.

Individual Cartridges Lower Costs

Nova's unique zero maintenance cartridge system consists of individual cartridges for sensors, calibrators, and liquid QC. This design optimizes the life of each cartridge, improves analyser uptime, eliminates waste, and reduces the high costs associated with combined cartridges. For example, an analyser used in a high volume setting requires fewer sensor cartridges than calibrators, and in a low volume setting the reverse. In both cases, costs are reduced by using fewer cartridges overall.

There is added savings in analyser uptime when only a calibration cartridge is replaced: No warm up time compared to the two to four hours required for a combined calibration/sensor cartridge. Reductions in downtime and costs of operation are often significant when compared to the inflexibility of older generation, combined systems.

Cartridge replacement in seconds

Each cartridge is easily replaced and ready to use in seconds. Cartridge RFID technology automatically captures installation time, date, lot number, test parameters, and usage.

Safe, zero maintenance waste system

Biohazardous waste is self-contained within the Calibration Cartridge, eliminating waste maintenance and potential user exposure.



Nova's MicroSensor cards, calibration cartridges, and quality control cartridges offer a zero-maintenance solution to critical care testing.

MicroSensor cards have fastest replacement time

MicroSensor cards can be replaced, warmed, calibrated, and ready to run the analyser's full sample throughput in less than half the time it takes other cartridge systems. These systems can take more than one hour to calibrate and can remain unstable with drift, frequent re-calibrations, and reduced throughput for even longer periods of time.

Safe Operation

A unique safety sample port protects the user from accidental contact with the sample probe, and is easily accessed by all sample containers.



Syringes can be docked and sampled with hands-free operation.



Capillary sampling can be performed without adapters.



Samples can be aspirated directly from tubes, eliminating transfer to a syringe or capillary.



Control proficiency ampoules can be sampled without adapters.

Automated, True Liquid Quality Control



QC statistics and reports are automatically maintained and easily accessed.

True liquid QC is the only reliable way to test an analyser

In the United States, federal government regulations (CLIA) are phasing out electronic equivalent QC and are requiring true, liquid-based QC. Stat Profile Prime combines both automated, true liquid QC and continuous electronic self-monitoring to ensure lab accuracy and uncompromised quality.

Tri-Level Quality Control Cartridge automates daily QC

The Quality Control Cartridge contains a 35-day supply of liquid QC material. Controls are run automatically at user selected intervals. This automated system complies with new U.S. CLIA requirements and other regulatory standards.

Saves time and labor

Maintaining QC is one of the most time consuming aspects of critical care testing. Stat Profile Prime’s automated, true liquid QC saves hours of time each week.

Supplemental Quality Monitoring (SQM)

Stat Profile Prime provides a supplement to liquid QC. SQM continuously monitors the status and performance of all analytical components (including sensors, calibrations, sample integrity, software, and electronics), providing real-time, sample-to-sample assurance of correct performance.

Compact, Point-of-Care Size

Stat Profile Prime’s micro-electronics and cartridge system result in one of the smallest and lightest critical care analysers. Stat Profile Prime is so compact it can be located virtually anywhere in the hospital or operated on a mobile cart with a battery back-up.



Stat Profile Prime Models

Five models offer a choice of test menus from basic electrolytes to a comprehensive, 10-test menu with blood gases, electrolytes, and metabolites.

Electrolyte Model



Basic Electrolyte Model

Na, K, Cl or Na, K, Cl, Li

Comprehensive Electrolyte Model

Na, K, Cl, iCa, iMg

Acceptable Samples

Whole blood (heparinized), arterial, mixed venous, capillary, serum, plasma, urine

Sample Volume

Basic Electrolyte Model 50 µl

Comprehensive Electrolyte Model 100 µl

Calculated Parameters

niCa, niMg, niCa/niMg

Critical Care Model



Blood Gas Model

pH, PCO₂, PO₂

Blood Gas/Electrolyte Model

pH, PCO₂, PO₂, Hct, Na, K, Cl, iCa

Blood Gas/Electrolyte/Metabolite Model

pH, PCO₂, PO₂, Hct, Na, K, Cl, iCa, Glu, Lac

Acceptable Samples

Whole blood (heparinized), arterial, mixed venous, capillary

Sample Volume

Blood Gas Model 50 µl

Blood Gas/Electrolyte Model 100 µl

Blood Gas/Electrolyte/
Metabolite Model 100 µl

Calculated Parameters

SO₂%, HCO₃⁻, TCO₂, Be-efc, Be-b, SBC, O₂Ct, O₂Cap, A, AaDO₂, a/A, RI, PO₂/FIO₂, Anion Gap*, P50*, Hb*

Temperature Corrected pH, PCO₂, PO₂

*Not available on the blood gas model

Optional Accessories

Combined 1D/2D Barcode Scanner

A barcode scanner is available for all Stat Profile Prime models. The scanner reads both 1D and 2D barcodes for patient and operator IDs. QC package inserts can also be scanned for lot number and expiration date.

Auto-Sampler

The Auto-Sampler is available for the Stat Profile Prime Electrolyte model. The 10 position tray accommodates serum, plasma, and urine samples in 2.0 ml and 0.5 ml sample cups. The Auto-Sampler is easy to load with prompting from Stat Profile Prime's user interface.

Stat Profile pHox Ultra Analysers

Nova's pHox Ultra analysers measure up to 20 tests from 210 ml of whole blood in only two minutes. The pHox Ultra test menu includes blood gases, SO₂%, hemoglobin, haematocrit, electrolytes including ionized calcium and ionized magnesium, glucose, lactate, urea/BUN, creatinine, and co-oximetry. Custom configured test menu models are available.



Measurement Range:

Na	80 - 200 mmol/L	Li	0.1 - 5.0 mmol/L	pH	6.50 - 8.00
K	1 - 20 mmol/L	Glu	0.8 - 28 mmol/L	PCO ₂	3.0 - 200 mmHg
Cl	50 - 200 mmol/L	Lac	0.3 - 20 mmol/L	PO ₂	0 - 800 mmHg
iCa	0.1 - 2.7 mmol/L	Hct	12% - 70%	BarP	400.0 - 800.0 mmHg
iMg	0.1 - 1.5 mmol/L				(53.3 - 106.7 kPa)

Certifications: ISO 9001 Quality System Registration, CSA, TÜV, CE Self Declared Complies to EN 61010, EN 50081,82

Physical Specifications:

Height: 39.06 cm (15.38 in) Width: 30.5 cm (12 in) Depth: 36.20 cm (14.35 in) Weight: 17.9 lb (8.167 kg) without calibration cartridge

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